To: Data & Analytics Team
From: Head of Retail Operations & Analytics
Date: October 22, 2026
Subject: Business Requirements Document (BRD) – E-Commerce Sales & Customer Insights Pipeline

1. Executive Summary

Over the past 24 months (June 2024 – June 2026), our e-commerce platform has processed approximately 250,000 transactions. While we have successfully modeled this data into our data warehouse's Gold Layer, leadership currently lacks visibility into key profitability drivers, customer retention metrics, and inventory turnover. We require an end-to-end analytical pipeline to transform these four Gold-tier tables (`customer`, `order`, `product`, `sales`) into an interactive executive dashboard that will guide our Q4 marketing and supply chain strategies.

2. Business Problem Statement

Our current reporting is highly fragmented. The marketing team cannot easily correlate customer tiers with specific coupon usage, and the supply chain team lacks insight into delivery SLA performance versus geographic regions. We need a unified, automated reporting solution to identify margin leakage, optimize regional pricing, and track the lifetime value of our customer base.

3. Project Scope

In-Scope:

- Exploratory Data Analysis (EDA) on the existing Gold Layer datasets to identify outliers, missing reviews, and stock anomalies.
- Development of key performance indicators (KPIs) relating to revenue growth, customer acquisition, and product performance.
- Design and deployment of an interactive, multi-page business intelligence dashboard.
- Documentation of all logic, calculations, and data models used.

Out-of-Scope:

- Modifying the Bronze or Silver ingestion layers.
- Real-time streaming analytics (batch processing of historical data is sufficient).
- Predictive machine learning models (focus strictly on descriptive and diagnostic analytics for this phase).

4. Data Assets Available (Gold Layer)

The project will utilize the following structured datasets:

Dataset	Key Fields	Description
`sales.csv`	`order_id`, `customer_id`, `product_id`, `quantity`, `unit_price`, `order_value`, `shipping_cost`, `total_amount`, `coupon_code`, `coupon_discount`	Transactional line items detailing item quantities, financials, and promotional metrics.
`order.csv`	`order_date`, `order_time`, `delivery_date`, `payment_mode`, `order_status`, `city`, `state`, `rating`, `review_text`	Order fulfillment records containing temporal data, logistics details, and customer feedback.
`product.csv`	`category`, `brand`, `price`, `discount_percent`, `selling_amount`, `stock_quantity`, `avg_rating`	Master catalog including product classification, pricing strategy, inventory, and sentiment metrics.
`customer.csv`	`city`, `state`, `pinecode`, `age_group`, `gender`, `tier`, `total_orders`, `total_spent`	Customer demographics and lifetime metrics including location, classification, and historical value.

5. Key Business Questions (The "Why")

Your analysis and final dashboard must directly answer the following questions for leadership:

Revenue & Profitability

- What is our total gross merchandise value (GMV) and net revenue after coupon discounts and shipping costs?
- Which product categories and specific brands are driving the highest margin versus the highest volume?

Customer Segmentation & Behavior

- How does purchasing behavior (average order value, category preference) differ across age groups and loyalty tiers (e.g., Gold vs. Platinum)?
- What is the geographical distribution of our highest-value customers?

Operations & Fulfillment

- What is the average delivery time (days between order and delivery), and does it vary significantly by state or payment mode (e.g., COD vs. Prepaid)?
- Is there a correlation between delivery times and customer ratings/reviews?

6. Technical Deliverables

As the lead analyst on this project, you are expected to deliver the following workflow:

1. **Data Validation (Python):** Use Pandas to conduct rigorous EDA. Document any data quality issues (e.g., missing delivery dates, invalid negative discounts) and provide a statistical summary of the operational metrics.
2. **Relational Modeling (SQL/Data Prep):** Ensure seamless primary/foreign key mapping across the four tables (e.g., joining `sales` to `product` via `product_id` and `sales` to `customer` via `customer_id`) using a strict star schema design.
3. **Executive Dashboard (Power BI):** Develop a 3-page interactive dashboard.
    - *Page 1: Executive Summary* (High-level KPIs, MoM growth, overall revenue).
    - *Page 2: Customer & Marketing Insights* (Tier breakdowns, coupon effectiveness, demographic slicers).
    - *Page 3: Product & Operations* (Inventory levels, category performance, delivery SLAs).
4. **Metric Calculations (DAX):** Implement robust DAX measures for Time Intelligence (YTD, MTD sales) and conditional metrics (e.g., Average Delivery Days).

7. Acceptance Criteria

- The final dashboard must allow users to filter dynamically by `order_date` (Month/Year), `category`, and `state`.
- Code and queries must follow first-principles logic—do not aggregate averages; calculate total revenue divided by total orders.
- Visualizations must be clean, adhering to corporate standards (avoiding cluttered pie charts in favor of bar charts, line graphs, and KPI cards).